

CoPH-SENSE: formulation and implementation progress

Research update relative to Prof. Bajaj’s revised manuscript

Implementation progress report

17 September 2026

Abstract

The revised manuscript asks when a scout should acquire and share terrain evidence so that a carrier can make a better *physically executable* decision under risk, resource, and communication constraints. We kept that causal ordering and the port-Hamiltonian (pH)/MHRF execution backbone. We first built exactly checkable finite tasks to validate set-valued sensing, post-observation sharing, persistent memory, and communication deadlines; then embodied those mechanisms in VMAS and integrated a fixed pH executor. The main remaining gap is a valid, frozen procedural benchmark in which physically acquiring complementary measurements is worthwhile often enough to train and test the full learner. E2-Natural is frozen and legitimately skip-heavy; the initial separate E2-Complementarity prototype has failed its physical admission rule. No held-out off-road or full E2 learning advantage is claimed here.

1 Starting point and research question

Prof. Bajaj’s updated Overleaf manuscript frames a *decentralized, risk-sensitive, pH-executed information decision*: a scout can learn about a future route before the carrier, but a measurement has value only if it can be obtained, communicated, and used before the carrier loses the ability to change maneuver. Sensing, detours, waiting, packet attempts, and both robots’ motion must be charged. The information policy is the new learning target; the pH/MHRF controller is the inherited structured physical backbone. Our implementation ladder tests the causal information mechanism before claiming benefits in continuous unknown-terrain transport. This follows the manuscript’s architecture rather than replacing it with a classification-only problem.¹

2 Notation, objective, and causal algorithm

For agent $i \in \{s, c\}$ (scout, carrier), x_t^i is its physical state and h_t^i its *local* history of measurements, actions, and packets actually delivered by time t . The latent map is $M = (O, g, \mu)$: public static occupancy O , hidden surface condition g , and hidden traction μ . Let $b_t^i = P(M, x_t \mid h_t^i)$ denote the ideal local belief (approximated in code), D a feasible set of sensing actions, Y_D their realized readings, and E a subset of *owned* evidence sent over the declared packet channel. The carrier chooses a route/maneuver ω ; the pH/MHRF layer produces an admissible force and executed control.

The manuscript’s full target is a decentralized, finite-horizon objective

$$\min_{\pi \in \Pi_{\text{dec}}} \sup_{P \in \mathfrak{P}} \left\{ \mathbb{E}_P[C_0(\pi)] + \lambda_R \text{CVaR}_\alpha^P(C_R(\pi)) \right\}, \quad \text{CVaR}_\alpha(Z) = \inf_{\eta} \left\{ \eta + \frac{\mathbb{E}(Z - \eta)_+}{1 - \alpha} \right\}, \quad (1)$$

subject to failure, sensing, radio, and actuator/safety constraints. C_0 includes both agents’ elapsed time/motion, sensing and communication charges, and failure penalties. C_R is cumulative material exposure. The current E2 nominal decision score sets $\alpha = .95$ and $\lambda_R = 1$ after fixed normalization:

$$S(\mathcal{I}, D) = \mathbb{E}[C_0 \mid \mathcal{I}, D] + \text{CVaR}_{.95}(C_R \mid \mathcal{I}, D), \quad C_R = \frac{\Delta t}{T_{\max}} \sum_{k < T} \sum_{i \in \{s, c\}} r_{\text{true}}(x_k^i), \quad (2)$$

where $r_{\text{true}} = \text{clip}(0.4g + 0.4t + 0.2gt, 0, 1)$ and $t = \text{clip}((0.55 - \mu)/0.40, 0, 1)$. A finite teacher rollout uses the same hidden world and random streams across actions, but the actor sees only its own h_t^i . The full ambiguity-set supremum and mission-level safety certificate in Eq. (1) are *not* established by the current VMAS experiments.

¹Source: revised `Distributed-Co-PH-Riskv2/CoPH_Sense_ICLR2027_Overleaf/`, especially `core.tex` and `learning_algorithm.tex`.

At a pre-acquisition information state $\mathcal{I}$, our restricted teacher evaluates complete sensing sets under a specified continuation:

$$Q_{\text{sense}}^{\pi}(\mathcal{I}, D) = \mathbb{E}[C_{\text{travel+dwel+sensor}} + C_{\text{packet}} + C_{\text{remaining}}^{\pi} \mid \mathcal{I}, \text{do}(D)], \quad D^* = \arg \min_{D \in \mathcal{D}(\mathcal{I})} Q_{\text{sense}}^{\pi}(\mathcal{I}, D). \quad (3)$$

After readings exist, $Q_{\text{share}}(\mathcal{I}^y, E)$ compares sending them with holding them under actual delay, range, and delivery. A positive interaction $Q_G + Q_T - Q_{GT} - Q_{\emptyset} > 0$ means the pair saves more team cost than the sum of its singleton improvements; it is *not* guaranteed by merely having two sensors. Acquisition regret always uses the same feasible menu and continuation. Exact evaluation of a fixed policy is not exact optimization over all decentralized policies.

Algorithmic sequence (current implementation and intended extension).

1. **Observe:** update each local belief/memory from its own sensors and delivered packets with age and provenance; keep hidden truth evaluator-only.
2. **Acquire (A3/A5):** enumerate feasible $\emptyset$, singleton and pair candidates; a variable-set critic predicts downstream mission cost and risk-return quantiles. Charge travel, dwell, sensing, and lost actionability.
3. **Share (A4):** after the reading is realized, compare SEND/HOLD for owned evidence. Requests do not reveal values; recipients update only on packet delivery. Persistent memory changes later candidate values.
4. **Act:** use the post-delivery risk belief to choose a maneuver and propose MHRF/pH motion. The full manuscript then applies its risk gate, energy tank, and safety filter; the present VMAS bridge uses a fixed structured executor, not the complete learned/certified controller.
5. **Train and check:** collect paired counterfactual returns on training worlds, fit A3/A5 and A4, refresh labels on histories visited by the learned stack, then evaluate complete decentralized rollouts. Freeze calibration and held-out manifests before confirmatory E2 claims.

The physical backbone follows the manuscript's $\dot{z} = [J(z) - R(z)]\nabla H(z) + Gu$ and its MHRF-style proposal

$$u_i^{\text{des}} = u_i^0 + \mathcal{A}_i \left[-g_i \lambda_i \nabla_{q_i} \bar{r}_i(q_i; \hat{b}_i) + f_i^{\text{coord}} \right]. \quad (4)$$

The proposed tank debits active-port power and storage jumps when the belief or controller changes. Our current result verifies integration with a fixed pH-style executor; it does not verify the entire tank/shield theorem in E2.

3 Why the finite checkpoints came first

The finite environments retain the manuscript's causal order while allowing exact comparison of sensing and sharing branches. They expose failures that a large navigation return could hide: leaked private evidence, a request revealing a reading, a message arriving after a maneuver deadline, an insufficient memory representation, or an optimistic A3 label that assumes oracle A4 sharing. The moving VMAS bridge then tests whether those decisions still matter when sensing and communication consume physical time. This is a verification ladder toward the intended pH unknown-terrain application, not a substitute for it.

Table 1: Measured milestones. Numbers from different tasks have different cost units and must not be compared across rows.

Stage	Measured result	Scope / reading
A0–A2	Native pH-MARL VMAS checkpoint: final assignment distance 0.103 vs 0.900 random (10 matched seeds). Finite oracle: 4 worlds, 16 sensing subsets, 625 sharing policies; best cost 2.6874 vs 3.0000 no sensing.	Upstream native checkpoint reproduced; finite oracle is an exactly enumerated diagnostic.
A3	Pairwise set critic: acquisition regret 0.00423 ± 0.00150 over three seeds (additive 0.01455 ± 0.00485).	Held-out finite teacher states; not E2 navigation improvement.
A4	Revised pairwise sharing critic: regret 0.01590; critical-message recall 99.33%, held-out warning recall 100%.	Post-reading finite SEND/HOLD; repaired state sufficiency and warning coverage.
A5	Learned memory-aware policy cost 5.2551, equal to restricted oracle; independent 5.6600, shared-reset 5.7006; nested regret 0.	Exact finite two-region task. Correct message changes the carrier’s next measurement from R1 to R2.
A6/A6.5	Finite timing gate: 97.36% decision agreement, regret 0.00182 vs 0.05885 always-inspect. Physical actionability gate: regret 0.0030 vs 0.0216 prior analytic VoI on a fresh 25-case set.	Deadline and moving single-fork sanity checks; the actionability correction is analytic, not a claimed learned contribution.

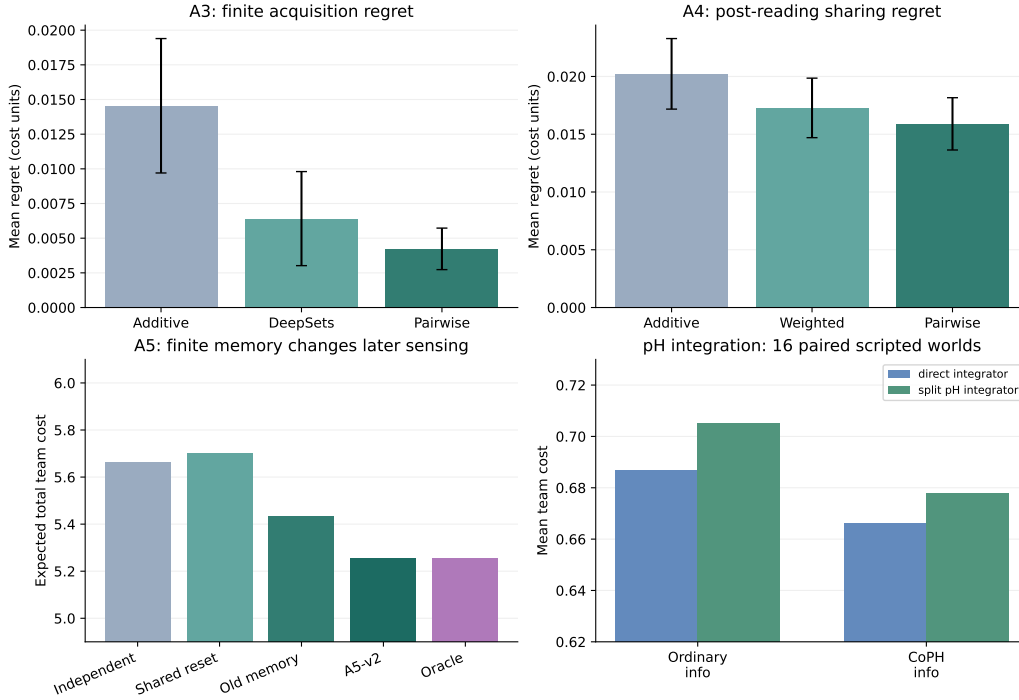


Figure 1: Finite diagnostics and scripted pH integration, recreated from saved result JSON. A3/A4 error bars are sample standard deviations across three seeds; A5 is the exact finite expected cost; the factorial averages 16 paired worlds. The four panels have distinct scales and objectives. The final panel compares two integrators of the *same* continuous-time pH vector field, not pH versus a structurally different non-pH controller.

4 Embodied bridge and pH execution

In the frozen two-fork scripted VMAS layout, the physical oracle chooses Region 1 without scout memory ($Q = 3.470$) and Region 2 after valid R1 evidence ($Q = 2.634$). The unchanged frozen A5 critic makes both choices, giving zero regret within this restricted menu. The matched safe-world physical cost decreases by 0.2866. An archived canonical rollout records actual scout G1/T1 acquisitions, two delivered packets, and carrier G2/T2 acquisitions; its completed team cost is 2.1163 versus 3.0894 for a shared-reset replay that repeats R1. These are particular scripted-VMAS branches, not generalization estimates.

The pH integration experiment holds the scripted worlds and information interface fixed while varying ordinary versus CoPH information and direct versus split numerical execution. Mean costs over 16 enumerated worlds are

	Ordinary information	CoPH information
Direct integrator	0.6871	0.6661
Split pH integrator	0.7051	0.6779

The information effect is -0.0209 under direct execution and -0.0272 under split execution, with 16/16 successes in every cell. Sensing cost (0.08) and communication cost (0.0488) are identical across cells, so this experiment shows altered downstream physical behavior, not reduced sensing or bytes. Since the two integrators share the same pH vector field, the split-minus-direct difference is a *numerical integration ablation*; it does not establish structural pH superiority.

5 Procedural unknown terrain: layouts, data, and current evidence

We then built a 12m $\times$ 12m VMAS scout-carrier world with public static topology, hidden surface and traction, passive noisy appearance, physical surface scans/contact traction probes, delayed charged packets, deterministic replay, and a complete resource ledger. The pH executor is kept fixed while the information policy is developed. The currently verified Natural manifests contain 500 train, 100 validation, 100 calibration, 200 in-distribution test, and 200 OOD parent groups; the test splits are not used for the engineering results here. RELLIS and TartanDrive remain planned data-derived evaluation sources, not completed E2 experiments.

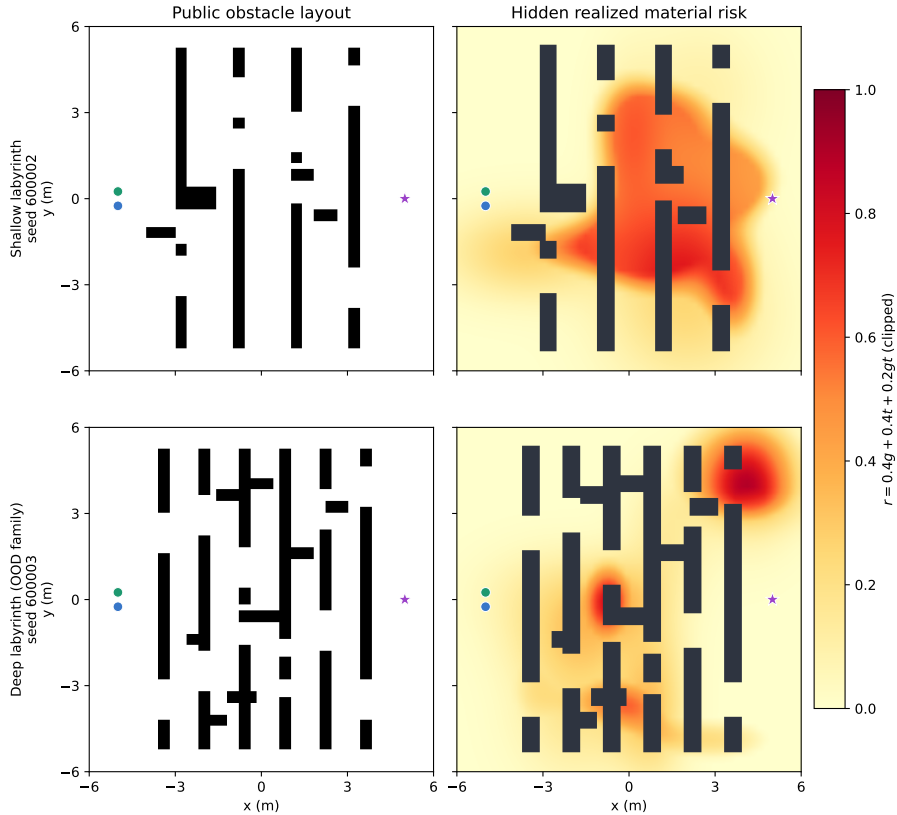


Figure 2: Current procedural shallow and deep labyrinth examples (seeds 600002/600003). Occupancy is public; colored material risk is hidden from actors and plotted from evaluator truth only. Green/blue dots are scout and carrier starts; purple stars are goals. Deep labyrinth is an OOD topology family. The colorbar uses the current joint surface-traction risk formula.

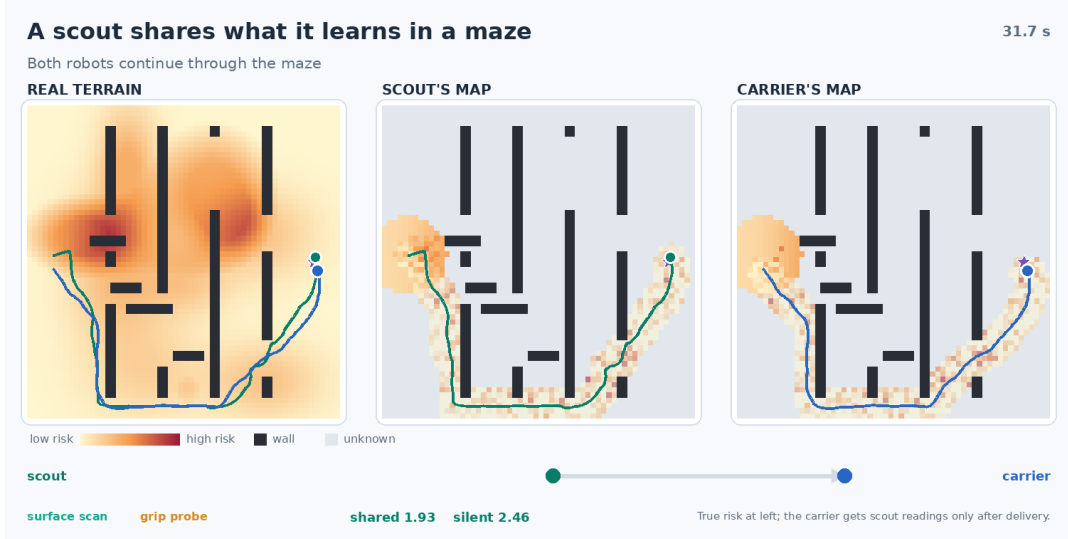


Figure 3: Qualitative information-flow replay in a labyrinth (parent seed 600042). Left is hidden true risk; middle and right are the separate local maps of scout and carrier. Scripted physical acquisition and two delivered packets make the carrier map update; with the same acquisitions, the paired send continuation costs 1.9321 versus 2.4563 when evidence is withheld. This illustrates the implemented causal path, *not* that buying the pair is optimal or that a learned E2 policy produced the trajectory.

6 Current blocker and next gate

The predeclared fresh E2-Natural resource audit covered all 40 scheduled states on 20 parent maps. Physical sensing beat skip by more than 0.01 in only 2/39 evaluated decisions (5.13%), below the 15% nondegeneracy gate; no pair was optimal. This is a valid result about the natural distribution: many measurements cost more in travel, dwell, and delay than the future decision benefit. We froze Natural instead of lowering prices or changing risk to force positive labels. Its purpose is now to test sensible restraint, success, cost, risk/CVaR, and resource use.

The separate controlled E2-Complementarity generator is still *provisional*. Admission uses an evaluator-only belief-expected physical oracle across indistinguishable latent worlds, requires every branch to succeed, both pair readings to be captured and delivered before region entry, and a pair advantage greater than 0.02 over skip and either singleton. It never screens on a CoPH checkpoint. The tested development parent failed:

Charged continuation	Skip	G	T	G+T	Outcome
Carrier moves	1.2814	1.3534	1.3511	1.4543	Pair loses; packets late
Carrier waits	1.2814	1.4159	1.4408	4.2692	One pair branch fails

In the high-surface/high-traction-risk state, the waiting pair branch costs 12.6835 and does not deliver both readings. These maps are therefore *not* pair-positive training examples. Disposable teacher forcing does execute an A4 SEND/HOLD comparison and a matched A5 retained/withheld state, but its tested safe-world A4 label favors HOLD (1.0439 versus 1.1064 SEND). Natural on-policy rollouts mostly skip, creating the expected coverage chain: skip $\Rightarrow$ no reading $\Rightarrow$ no A4 state $\Rightarrow$ no delivered evidence $\Rightarrow$ no meaningful A5 state.

Next gate. Specify and freeze a separate physically feasible Complementarity source pool, reporting its acceptance rate before fitting. Then teacher-force *training-only* acquisition to create real post-reading A4 states and paired post-delivery A5 states; fit the shared A3/A5 set critic and A4 critic on Natural-plus-challenge training states; refresh on histories generated by the learned stack. Held-out regret for A3/A4/A5 and a complete causal rollout must precede the matched rivals and multi-seed campaign. The pH executor remains fixed through this information learning step; optional pH proposal fine-tuning comes afterward. There is currently no completed held-out E2-CoPH result.

Reproducibility. This document, its figure generator, and source JSON references are in `reports/prof_progress_2026-09-17/`. The figures are regenerated from saved finite/VMAS results and current terrain generator outputs.